

Huahao Shang

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EDUCATION

Rice University

Master of Computer Science (GPA: 3.8/4.0)

Houston, TX

Jan 2023 - Dec 2024

Syracuse University

B.Sc. in Computer Science (GPA: 3.7/4.0. Dean's List)

Syracuse, NY

Jan 2019 - Dec 2022

PROGRAMMING SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C/C++

Backend: Node.js, Express.js, Django, Redis, RESTfulAPI, GraphQL

Frontend: React, HTML5, CSS, TailWind, Bootstrap, Next.js, Ajax

Database: MySQL, AzureSQL, MongoDB **Others:** PyTorch, Pandas, Jupiter, OpenGL

Tools: Git, VScode, Linux, Docker, Anaconda, AWS, GCP, Postman, Bash, OOP, Agile

WORK EXPERIENCE

Software Engineer Intern

Remote

LEARFIELD

May - Aug 2024

- Developed a data capture tool for collecting and analyzing live user engagement metrics during college sports streams. Integrated the tool into the **LEARFIELD Fanbase** dashboard to provide analytics insights for fan engagement.
- Leveraged **WebSockets** and **Redis** for persistent connections and efficient data caching, enabling real-time data capture and quick access for analytics
- Designed and implemented **GraphQL APIs** using **Node.js**, enabling optimized data fetching and dynamic querying for the dashboard
- Leveraged **MongoDB** for high-volume data storage, built a responsive **React** frontend, styled with **TailwindCSS**, to display real-time fan engagement metrics.
- Conducted comprehensive QA testing using **Jest** and **React Testing Library**, achieving **85%** code coverage, ensuring WebSocket connections and the proper behavior of real-time data updates.
- Collaborated with the Infrastructure team to set up an automated **CI/CD** pipeline using **Bitbucket Pipelines**. Integrated with **Jira** for workflow management and continuous feedback throughout the development cycle.

Web Developer Intern

Syracuse, NY

SIDEARM Sports

May - Aug 2022 and May - Aug 2023

- Developed web solutions for 30+ collegiate clients, including high-profile projects for **OSU** and **FSU**, responsible for **data imports**, **custom web component**, quality assurance and dedicate to **SIDEARM NextGen** product integration.
- Built customized **website components** using modern frontend technologies and integrated them with **SIDEARM Narrator**, ensuring flexible and scalable development for diverse client needs.
- Optimized **complex SQL queries** for data ingestion, leveraging **AWS S3** for efficient data storage, reducing game schedule import time from 1 hour to 30 minutes per sport.
- Conducted **data scraping and cleaning** to import athletic data into the client's **Content Management System (CMS)**, ensuring data integrity and accuracy.
- Defined and executed customized **QA** procedures, including, **usability testing**, and **functional testing**, ensuring seamless website performance throughout the development pipeline and post-launch.
- Reinforced **Agile software development lifecycle (SDLC)** practices, breaking down large projects into manageable milestones, driving on-time project delivery and successful website launches.

PROJECTS

Stream & Connecting For Yall

Oct 2023 - Feb 2024

- Developed a full-stack web enabling users to **live stream** and build fans community.
- Implemented server-side logic with **Next.js** and **RESTful API**; built frontend with **React** and **TailwindCSS**.
- Integrated **Prisma ORM** with **MySQL** for database managing to ensure effective data operations.
- Built a streaming service with **LiveKit** and **Ingress**, and incorporated live chat using **Socket.io**. Utilized **Redis caching** to store live chat data, reducing database load and response time.
- Implemented versatile features to enhance platform engagement and user retention, including fan management, push notifications, and personalized user experiences.

Web Social

Sept - Nov 2023

- Developed a full-stack social media web app, designed dynamic frontend interface using **React** and **Bootstrap**.
- Built a resilient backend server with **Node.js**, **Express.js**, leveraging **RESTful API** with **MongoDB**, ensuring data storage, error handling and efficient asynchronous operations.
- Implemented **CORS** settings to achieve seamless Ajax requests between separated frontend and backend systems.
- Deployed the app on **Surge** (frontend) and **Heroku** (backend), conducted component testing with **Jest** and **React Testing**, achieving **90%** code convergence.
- Designed an intuitive UI, enabling CRUD operations for posts, image handling, following system, and integrated third-party authentication using **GoogleOauth**.

Solar System Simulation

Feb - May 2022

- Built a graphical simulator in **C++** using **OpenGL**. Enabling interactive animation of the solar system in detail.
- Designed **Object-Oriented** models and functions for general planet objects and compound planet systems.
- Designed a random **Particle System** for vivid visual of asteroid belts with high dense.
- Implemented lighting techniques and vector calculations to better simulate solar light and shadows.
- Used advanced OpenGL topics including glut objects, **Bézier surfaces**, textures for advanced simulation.

Pac-Man Game Agent

Sept - Dec 2021

- Developed a sophisticated Pac-Man agent utilizing machine learning techniques, aimed at path optimization and maximizing game scores in a dynamic multi-agent environment in Python.
- Utilized **Reinforcement Learning** (Q-Learning) along with classic algorithms like DFS, BFS, and **Dijkstra's** to enhance the agent's pathfinding and decision-making capabilities.
- Achieved recognition with the agent being ranked within the top five for intelligence and game strategy effectiveness in competitive evaluations.

EXTRA-CURRICULUM

Rice Computer Science Club

Feb 2023 - Current

- Hosted workshops to demo and present various algorithms.
- Participated in coding challenges, focusing on algorithm problem-solving.

Player & Referee, Chinese Basketball Team at SU

Feb 2021 - Dec 2022

- Played and competed in NECBL (Northeastern Chinese Basketball League) for Syracuse University.
- Contributed to team operation and SU student basketball development. Refereed for competitions for 40+ games.

Recruiting Assistant, Dept. of Engineering and Computer Science

Jan 2020 - Dec 2022

- Led 10+ discussion seminars with high school students and families to introduce the college life at SU and FAQ.
- Planned and hosted 5+ engagement events to help potential future engineering students to know our department better.